

Free Response

Write your response to the following question on the back of the response sheet. Show your work for partial credit. **Box your final answer for each part.**

28. Starlight Cinemas is the only movie theater in town. They face an inverse demand curve for movie tickets given by $P = 10 - 0.5Q$, which implies a marginal revenue curve of $MR = 10 - Q$. The theater's marginal cost for each additional ticket sold is constant at $MC = 4$. Assume the theater has no fixed costs.
- Assuming Starlight Cinemas charges a single price to all customers, what is the profit-maximizing quantity (Q^*) and price (P^*)? Show your work and box your final answer.
 - Calculate the deadweight loss (DWL) in the single-price scenario. Show your work and box your final answer.
 - Calculate the theater's total profit at the single price. Show work; box answer.
 - Now suppose the theater's demand is actually made up of two distinct groups:
 - Locals, whose demand is given by $P_L = 12 - Q_L$ (implying $MR_L = 12 - 2Q_L$), and
 - Visitors, whose demand is given by $P_V = 8 - Q_V$ (implying $MR_V = 8 - 2Q_V$).
 - Note: Adding these two demands together yields the exact "aggregate" demand given initially.*If the firm *cannot* distinguish between Locals and Visitors, what price should it charge? How does it compare to the single-price implied by the above parts? Explain briefly. Box your answer.
 - If the firm *can* successfully distinguish between the two types of customers (e.g., by checking for a local zip code on their ID) and prevent resale, what price will they charge each group, and what is the new total profit they could achieve? Show work; box answer.
 - If the firm can distinguish between the two types of customers, does the *efficient price* (i.e., the price that would lead to an efficient outcome) change? Briefly explain why or why not.